

Q & Soergel 10 : Soergel theory 2

- 1) Completion of proof of full faithfulness
- 2) Deformed Soergel functor

1) Completion of proof of full faithfulness

1.0) Recap from Lec 9

Let $\Xi = W \cdot \lambda \in \Lambda$ & λ^+, λ^- be the unique elements of Ξ s.t. $\lambda^+ + \rho$ is dominant & $\lambda^- + \rho$ is antidominant. We get the projective $P(\lambda^-) \simeq \mathcal{T}_{-\rho \rightarrow \lambda^+} \Delta(-\rho)$ (Sec 2.2 of Lec 9)

We have the functor

$$\mathbb{V}_\lambda := \text{Hom}_{\mathcal{O}}(P(\lambda^-), ?) : \mathcal{O}^\Xi \longrightarrow \text{End}_{\mathcal{O}}(P(\lambda^-))^{\text{opp}}\text{-mod}_{fg}$$

In Lec 9 we've stated that $\mathbb{V}_\lambda$ is fully faithful on projectives and reduced this to the existence of a certain exact sequence. Namely, recall the projective generator $\tilde{P} = \mathcal{T}_{\lambda' \rightarrow \lambda^+} (\bigoplus_{w \in W} \bigoplus_{\underline{w}}) \Delta(\lambda')$ of $\mathcal{O}_\Xi$, where λ' & $\lambda' - \lambda^+$ are dominant. In this section, we prove:

1 Proposition: $\exists$ exact sequence $0 \rightarrow \tilde{P} \rightarrow P(\lambda^-)^{\oplus m} \rightarrow P(\lambda^-)^{\oplus n}$

1.1) $P(\lambda^-)$ vs translation functors.

$\tilde{P}$ is obtained from $\Delta(\lambda')$ by applying projective functors & taking $\oplus$'s. So it's a natural question to ask what $\mathcal{T}_{\lambda \rightarrow \lambda'}$ do to $P(\lambda^-)$. Recall that the $\mathcal{T}$ -functors are defined in the following setting: we pick $\lambda_1, \lambda_2 \in \Lambda$ s.t. $\lambda_1 + \rho$ & $\lambda_1 - \lambda_2$ are dominant

Proposition: 1) $\mathcal{T}_{\lambda_2 \rightarrow \lambda_1} P(\lambda_2^-) \cong P(\lambda_1^-)$
2) $\mathcal{T}_{\lambda_1 \rightarrow \lambda_2} P(\lambda_1^-) \cong P(\lambda_2^-)^{\oplus r}$, where
 $r := |\text{Stab}_W(\lambda_1 + \rho) / \text{Stab}_W(\lambda_2 + \rho)|$

Proof: we deduce these claims from the corresponding parts of the theorem in Sec 1 of Lec 7 (that describes $\mathcal{T}_{\cdot \rightarrow \cdot} \Delta(-)$) & Sec 2.2 of Lec 9. We'll prove the harder part - 2) - & leave 1) as **exercise**.

By 2) of Prop'n in Sec 2.2 of Lec 9, it's enough to prove: $\dim \text{Hom}_\mathcal{O}(\mathcal{T}_{\lambda_1 \rightarrow \lambda_2} P(\lambda_1^-), \Delta(\lambda)) = r \quad \forall \lambda \in W \cdot \lambda_2$
l.h.s = $\dim \text{Hom}_\mathcal{O}(P(\lambda_1^-), \mathcal{T}_{\lambda_2 \rightarrow \lambda_1} \Delta(\lambda)) = [\mathcal{T}_{\lambda_2 \rightarrow \lambda_1} \Delta(\lambda) \text{ is an iterated extension of } r \text{ Verma's \& } \text{Hom}_\mathcal{O}(P(\lambda_1^-), \Delta(\mu)) \cong \mathbb{K}$

2] $\forall \mu \in W \cdot \lambda_1] = r.$

□

1.2) Proof of Proposition in Sec 1.0.

Thx to Sec 1.1 & the exactness of J 's & $\oplus$'s it's enough to establish an exact sequence

$$0 \rightarrow \Delta(\lambda') \rightarrow P(\lambda') \rightarrow P(\lambda')^{\oplus n}$$

This is Lemma in Sec 2.4 of Lec 9.

2) Deformed Soergel functor

2.1) Discussion

We take $A := \mathbb{C}[[\mathfrak{h}^*]]$ w. the natural map

$$\chi: S(\mathfrak{h}) = \mathbb{C}[\mathfrak{h}^*] \hookrightarrow A \text{ so that } \chi \in \mathfrak{h}^* \otimes m.$$

Pick $\lambda \in \Lambda$ s.t $\lambda + \rho$ is dominant. Recall (Example in Sec 2.1 of Lec 8) that

$$\text{Hom}_{\mathcal{O}_A}(\Delta_A(-\rho), ?): \mathcal{O}_A^{\{-\rho\}} \xrightarrow{\sim} A\text{-mod}_{fg}$$

$$\text{Define } \hat{V}_\lambda: \mathcal{O}_A^{W \cdot \lambda} \rightarrow A\text{-mod}_{fg} \text{ as}$$

$$\hat{V}_\lambda(?) = \text{Hom}_{\mathcal{O}_A}(\Delta_A(-\rho), T_{\lambda \rightarrow -\rho}(?))$$

The action of W on $\mathfrak{h}$ induces that on A by algebra automorphisms.

3] Write $W_o = \text{Stab}_W(\lambda + \rho)$. We'll construct a functor

$$\hat{V}_\lambda: \mathcal{O}_A^{W, \lambda} \longrightarrow A^{W_0} \otimes_{A^W} A\text{-mod}_{fg}$$

so that $\underline{V}_\lambda$ is the composition of V_λ & the pullback under the inclusion $A \hookrightarrow A^{W_0} \otimes_{A^W} A$, $a \mapsto 1 \otimes a$.

2.2) Center of a category

Definition: Let $\mathcal{C}$ be an A -linear abelian category. By the center of $\mathcal{C}$, $Z(\mathcal{C})$, we mean the A -algebra of the endomorphisms of the identity endofunctor of $\mathcal{C}$.

Example: Let E be an (associative unital) A -algebra. Take $\mathcal{C} := E\text{-Mod}$ (if $E\text{-mod}_{fg}$ is E is Noetherian). We claim that $Z(\mathcal{C})$ is isomorphic to $Z(E)$, the center of E . Namely we send $\varphi \in Z(\mathcal{C})$ to $\varphi_E \in \text{End}_E(E) \simeq E^{\text{opp}}$. Since φ is a functor endomorphism, φ_E is central. Conversely, $Z(E)$ acts by E -linear endomorphisms on any E -module giving a homomorphism $Z(E) \rightarrow Z(\mathcal{C})$. To show these two homomorphisms are mutually inverse is

4 left as an exercise.

Note that if $\mathcal{C} \subset E\text{-Mod}$, then we still have an A -algebra homomorphism $Z(E) \rightarrow Z(\mathcal{C})$ but it may fail to be injective or surjective.

The reason why we need the center of $\mathcal{C}$ is that it acts on any A -linear functor F from $\mathcal{C}$ thx to a functor isomorphism $F \xrightarrow{\sim} F \circ \text{id}_{\mathcal{C}}$.

2.3) Construction of $\hat{V}_\lambda$, I.

In particular, $\underline{V}_\lambda$ lifts to a functor

$$\mathcal{O}_A^{W \cdot \lambda} \longrightarrow Z(\mathcal{O}_A^{W \cdot \lambda})\text{-mod}_{\mathfrak{g}} \quad (1)$$

(where "lifts" means that $\underline{V}_\lambda$ is the composition of that with the pullback under $A \rightarrow Z(\mathcal{O}_A^{W \cdot \lambda})$). We are going to construct an A -algebra homomorphism (that will later be shown to be an isomorphism)

$$A^{W_0} \otimes_{A^W} A \longrightarrow Z(\mathcal{O}_A^{W \cdot \lambda})$$

The construction is a version of the HC isomorphism:

5 $Z \xrightarrow{\sim} \mathbb{C}[\mathfrak{g}^*]^{(W, \cdot)}$, where Z is the center of $U(\mathfrak{g})$. By

Example in Sec 2.2, we have an A -algebra homomorphism

$$\mathbb{Z} \otimes A \xrightarrow{\text{in fact } =} \mathbb{Z}(\mathcal{U}(\mathfrak{g}) \otimes A\text{-mod}) \longrightarrow \mathbb{Z}(\mathcal{O}_A^{W \cdot \lambda})$$

For $\mu \in \mathfrak{h}^*$ consider the affine shift isomorphism

$$\mathfrak{h}^* \xrightarrow{\zeta_\mu} \mathfrak{h}^*, \quad x \mapsto x - \mu.$$

The pullback $\zeta_{\lambda^+}^*: \mathbb{C}[\mathfrak{h}^*] \xrightarrow{\sim} \mathbb{C}[\mathfrak{h}^*]$, $x \in \mathfrak{h} \mapsto x + \langle \lambda^+, x \rangle$ restricts to $\mathbb{C}[\mathfrak{h}^*]^{(W_0, \cdot)} \xrightarrow{\sim} \mathbb{C}[\mathfrak{h}^*]^{W_0}$ (exercise) yielding

$$\mathbb{Z} \xrightarrow{\sim} \mathbb{C}[\mathfrak{h}^*]^{(W_0, \cdot)} \xrightarrow{\sim} \mathbb{C}[\mathfrak{h}^*]^{W_0} \hookrightarrow A^{W_0} \xrightarrow{\sim} \mathbb{Z} \otimes A \hookrightarrow A^{W_0} \otimes A$$

Exercise: The composition

$$\mathbb{Z} \otimes A \hookrightarrow A^{W_0} \otimes A \longrightarrow A^{W_0} \otimes_{A^W} A$$

is surjective (hint: show that

$$(\mathbb{C}[\mathfrak{h}^*]^{W_0} \otimes_{\mathbb{C}[\mathfrak{h}^*]^{W_0}} \mathbb{C}[\mathfrak{h}^*]) \otimes_{\mathbb{C}[\mathfrak{h}^*]} A \xrightarrow{\sim} A^{W_0} \otimes_{A^W} A)$$

Proposition: The homomorphism $\mathbb{Z} \otimes A \longrightarrow \mathbb{Z}(\mathcal{O}_A^{W \cdot \lambda})$

factors through $\mathbb{Z} \otimes A \longrightarrow A^{W_0} \otimes_{A^W} A \hookrightarrow \mathbb{Z}(\mathcal{O}_A^{W \cdot \lambda})$

The proof requires a key computation (Example below)

6] & some formalism. We'll finish it in Lec 11.

Example: For $w \in W$ we have the algebra homomorphism $\mathbb{Z}(\mathcal{O}_A^{w \cdot \lambda}) \xrightarrow{\varphi \mapsto \varphi_{\Delta_A(w \cdot \lambda^+)}} \text{End}_{\mathcal{O}_A}(\Delta_A(w \cdot \lambda^+)) \xrightarrow{\sim} A$. We want to compute its composition with $\mathbb{Z} \rightarrow \mathbb{Z}(\mathcal{O}_A^{w \cdot \lambda})$.

$\mathbb{Z}$ acts on $\Delta_A(w \cdot \lambda^+)$ via (exercise on construction of HC):

$$\varphi_w: \mathbb{Z} \xrightarrow[\text{HC.}]{\sim} \mathbb{C}[\mathfrak{h}^*]^{(W, \cdot)} \subset S(\mathfrak{h}) \xleftarrow{x \mapsto \langle w \cdot \lambda^+, x \rangle + X(x) (x \in \mathfrak{h})} A.$$

where, recall X stands for the natural inclusion $S(\mathfrak{h}) \simeq \mathbb{C}[\mathfrak{h}^*] \hookrightarrow \mathbb{C}[[\mathfrak{h}^*]]$. In other words, $\varphi_w(z)$ is the evaluation of $\text{HC}_z \in \mathbb{C}[\mathfrak{h}^*]^{(W, \cdot)}$ at $w \cdot \lambda^+ + X \in \mathfrak{h}^* \otimes A$. Because of the $(W, \cdot)$ -invariance this is the same as evaluating at $w^{-1}(w \cdot \lambda^+ + X) = \lambda^+ + w^{-1}X$, i.e. the evaluation of $\zeta_{\lambda^+}^* \text{HC}_z \in \mathbb{C}[\mathfrak{h}^*]^{W_0}$ at $w^{-1}X$, i.e. the evaluation of $w \zeta_{\lambda^+}^* \text{HC}_z \in \mathbb{C}[\mathfrak{h}^*]$ at X . So

$$\varphi_w(z) = X(w \zeta_{\lambda^+}^* \text{HC}_z) \in A$$

In particular, φ_w factors as the composition

$$\mathbb{Z} \xrightarrow{z \mapsto \zeta_{\lambda^+}^*(\text{HC}_z)} A^{W_0} \subset A \xrightarrow{w} A$$

The resulting homomorphism $A^{W_0} \otimes A \rightarrow A$, $(b, a) \mapsto w(b)a$, indeed factors through $A^{W_0} \otimes_{A^w} A$ (exercise).

2.4) Localization vs Homs

The proof of Proposition uses localization.

Let B be a local ring that is a localization of A .

Two choices we are going to use:

0) $B := \text{Frac}(A)$

1) $B := A_{\mathfrak{p}}$ for a minimal (height 1) nonzero prime ideal $\mathfrak{p}$.

Exercise: $B \otimes_A \cdot$ is an exact functor $\mathcal{O}_A \rightarrow \mathcal{O}_B$
(where B is an $S(\mathcal{Y})$ -algebra via $S(\mathcal{Y}) \xrightarrow{x} A \rightarrow B$)

To simplify the notation, for $M \in \mathcal{O}_A$, set $M_B := B \otimes_A M$.

Lemma: 1) $B \otimes_A \text{Hom}_{\mathcal{O}_A}(M, M') \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_B}(M_B, M'_B)$, $\forall M, M' \in \mathcal{O}_A$.
2) If $P \in \mathcal{O}_A$ is projective, then P_B is projective in $\mathcal{O}_B$.

Sketch of proof (details are **exercise**)

1) Follows b/c M is finitely presented over $\mathcal{U}(\mathcal{g}) \otimes A$ (as a fin. gen'd module over a Noetherian ring). 2) follows

b/c of 1) & $B \otimes_A \cdot: \mathcal{O}_A \rightarrow \mathcal{O}_B$ being essentially surjective. $\square$

2.5) Localization vs Verma filtration.

Here we consider B of one of the following forms:

- (i) $B = \text{Frac}(A)$, or
- (ii) $B = A_{\mathfrak{p}}$ where $\mathfrak{p}$ is a minimal prime different from (d^v)

Here's the main reason why we care

Proposition: Let $M \in \mathcal{O}_A$ has Verma filtration:

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \dots \subsetneq M_r \text{ w. } M_i/M_{i-1} \simeq \Delta_A(\lambda_i).$$

For B as above, the resulting filtration of M_B splits, i.e. $M_B \simeq \bigoplus_{i=1}^r \Delta_B(\lambda_i)$.